

Learning Deep Multimodal Latent Representations from Wearable and Sensor Data

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ABSTRACT

The growing number of wearables and ubiquitous computing devices has provided access to data of various modalities on-the-fly. This data is used for multiple applications, such as health-sensing, human activity and context recognition, dead-reckoning, and navigation. Integrating data from various devices and sensing modalities requires task-specific feature selection and processing on data of different modalities independently before fusing them together. Deep Learning provides an alternative by allowing us to fuse multiple modalities to create compact shared latent representations which are usable across tasks. This research proposal is focused on determining the best methods to create deep latent representations from multimodal data, from the perspectives of quality of features, their versatility across tasks, and related issues such a method can address, like missing data, compression, privacy preservation, and sensor fluctuation.

1 INTRODUCTION & OBJECTIVES

Multimodal data from wearables and edge devices is being used in various contexts, from HAR, to biosensing, to pedestrian dead reckoning (Mukhopadhyay, 2015; Hou and Bergmann, 2020; Ding et al., 2020). As the scale of this data increases, traditional algorithms may not be able to handle its variability and dimensionality. Combinations of sensing modalities, for example, IMU sensors, Wifi fingerprinting, GPS data, and audio-visual data, can lead to increase in the accuracy of task; however, manual processing and fusion of each individual modality is cumbersome and varies from task to task. Moreover, most existing approaches do not take into account heterogeneity of devices and the fact that not all modalities may be available at all times. As such, there is no way to combine and represent multimodal sensor data in a compact way that can be used across tasks.

In previous work in the context of medical imaging, we show that deep neural networks learn disentangled representations of relevant concepts from input data (Natekar et al., 2019). In the context of wearable and sensor data as well, deep learning has the potential to provide a solution to the issues discussed above by allowing us to generate combined latent space representations of multiple sources of data with varying levels of abstraction that learn the underlying factors of variation of the data. This latent space representation may then be used for various tasks, such as supervised classification or unsupervised clustering. This proposal focuses on two directions: (1) Determining the best methods and architectures to generate deep multimodal latent representations from heterogeneous wearable, visual, and sensor data, so that they can be used across tasks without individual processing for each task, and (2) How such deep latent representation

based methods could address related issues in this context such as compression, incomplete and fluctuating sensor data, synthetic generation, and privacy preservation.

2 SCOPE AND METHODS

2.1 Methods for Multimodal Deep Learning for Wearable and Sensor Data

General purpose deep learning techniques have been shown to perform better than than purpose specific shallow machine learning techniques (SVMs, Decision Trees) over a range of multimodal detection tasks (Radu et al., 2018). The performance increase that deep learning offers is due to the power of automatically learnt deep latent representations. A number of methods have been proposed for creating such deep multimodal representations. Radu et al. (2018) compare feature concatenation and a modality-specific architecture which consist of sensor-specific hidden layers followed by a common fused hidden layer. Ngiam et al. (2011) propose multimodal autoencoder structures that have the potential not only to create multimodal latent representations but also to reconstruct complete representations when only some modalities are available.

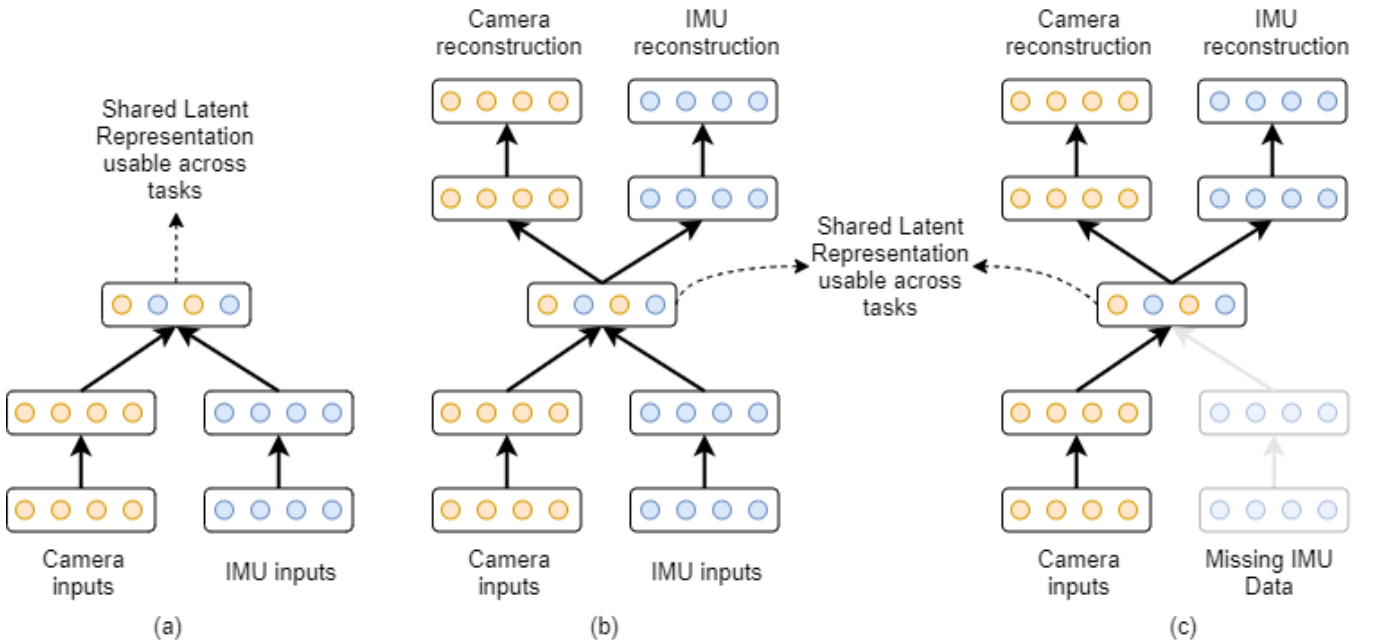


Figure 1. (a) Joint Representation Learning for IMU and visual stream, (b) Multimodal Autoencoder for IMU sensor data and visual data. The shared latent representation is highly expressive and could be used for multiple tasks without requiring individual processing for each task, (c) Reconstructing data with missing modalities. While I have shown IMU and video data, any set of multimodal inputs could be modelled as above.

I borrow from Guo et al. (2019) to divide methods of multimodal deep learning frameworks into the following: (i) Joint or Co-ordinate representation learning, which aims to learn a shared or correlated subspace by either combining modalities at the input or a hidden layer, or by constraining representations of different modalities, and (ii) Encoder-Decoder models, for example, multi-modal autoencoders which create a latent representation by learning to fuse and then reconstruct the different input modalities.

Multimodal variational autoencoders have been explored before for modality exchange or joint learning, for example converting text to images or reconstructing video and audio when only one modality is available (Suzuki et al., 2016; Ngiam et al., 2011). Figure 1(b) shows a potential multimodal autoencoder

architecture that learns a combined representation from IMU and visual data. Mathematically, the aim is to find an intermediate representation $\phi(X, W)$, where X and W are two different input modalities:

$$\phi(X_i, W_i) = \mathcal{E}(X_i, W_i) \quad (1)$$

$$\hat{X}_i, \hat{W}_i = \mathcal{D}(\phi(X_i, W_i)) \quad (2)$$

Such that the reconstruction loss is minimized:

$$\mathcal{L}_{AE} = \sum_{i=1}^n \|\hat{X}_i - X_i\|_2^2 + \|\hat{W}_i - W_i\|_2^2 \quad (3)$$

Here, X_i and W_i are two input samples for the two different modalities, n is the total number of input samples, $\mathcal{E}$ is the encoding section and $\mathcal{D}$ is the decoding section of the autoencoder. For VAEs, an additional KL divergence term is added that encourages the learnt proxy distribution $q_j(Z|X, W)$ to be closer to the true prior $p(Z)$:

$$\mathcal{L}_{VAE} := \mathcal{L}_{AE} + \sum_{j=1}^m KL(q_j(Z|X, W)||p(Z)) \quad (4)$$

Where m is the number of dimensions in the intermediate representation.

Such multimodal autoencoders and VAEs could be leveraged to learn shared feature representations $\phi(X_1, X_2 \dots X_k)$ that encode higher order correlations across modalities. For instance, complex relationships between an audio-visual stream from a head-mounted camera and accelerometer and gyroscope data from an IMU could be captured in a compact fashion. This shared representation can then be picked up as is and used for any task such as activity recognition, dead reckoning, sleep stage detection etc.

In the context of wearable and sensor data, the requirements of such deep multimodal representations would be: (i) Able to handle the heterogeneity and fluctuation of wearable and sensor data, (ii) Can exploit a varying set of modalities that are available at any given time to generate consistent representations, (iii) Usable for multiple applications without task-specific pre-processing.

This proposal aims to evaluate and compare multimodal autoencoders, joint learning approaches, and other multimodal learning frameworks from the perspective of the three requirements detailed above and compare them with existing shallow approaches. It also aims to look at other considerations in this context - for instance, corrupted or missing data, data compression, and generalization.

In the next sections, I discuss the potential of such methods to address some common issues in the context of wearable and sensor.

2.2 Dealing with Missing Modalities and Generating Synthetic Data

In a wearable or ubiquitous computing setting, corrupted or missing data is common. Multimodal deep learning algorithms provide the capability to reconstruct the shared representation in the absence of some modalities. For example, (Ngiam et al., 2011) detail a hybrid training method for encoder-decoder architectures that allows the model to reconstruct both video and audio when the audio inputs are missing. Thus, a shared generative representation using, for instance, a multimodal VAE, would allow us to deal with

the situation of missing data, as shown in Figure 2(b). This would be especially helpful in our context, and would allow, for instance, reconstructing both IMU and camera streams when only one of the modalities is available.

Another use-case of such a capability would be in creating synthetic multimodal data which closely resembles the input dataset. This may help in resource constrained settings where data is scarce. More complex architectures like generative-adversarial networks has also be used for synthetic generation in the context of HAR (Wang et al., 2018; Soleimani and Nazerfard, 2019), and is another potential direction for research.

2.3 Data Compression and Privacy

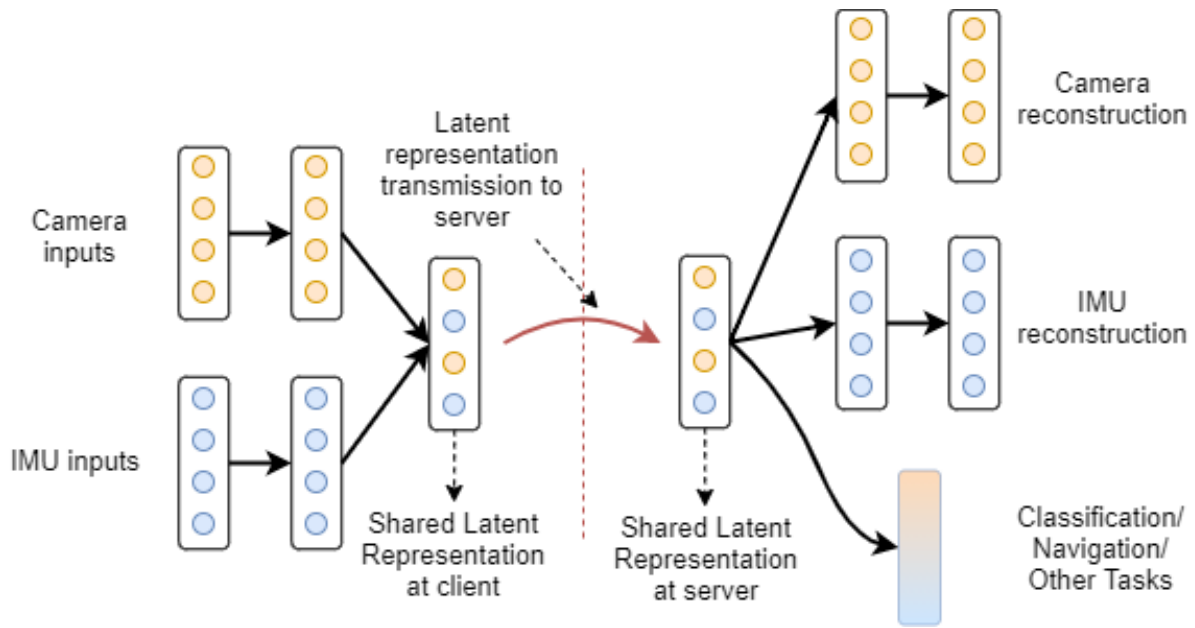


Figure 2. A split multimodal representation learning approach for compression and privacy preservation.

Data compression and privacy are both important requirements in the context of wearable computing and m-health applications. Shared latent representations have the potential to add value in both these directions. Autoencoders have traditionally been used for image compression (Mukherjee and Chandran, 2017). Generating a compressed latent representation before transmitting data and reconstructing it at the server side using a split autoencoder has been used before for m-health applications (Al-Marridi et al., 2018). From the perspective of privacy, Vepakomma et al. (2018) use a split learning approach for collaborative training where only the latent representations are shared between entities. Deep multimodal representations could thus provide an integrated approach for both compression and privacy preservation. Figure 2 shows such a split-learning approach where only the compact latent representations are shared between the client and server. This ensures that raw sensor values are not shared between the client and server, and only the compressed latent representation need to be transmitted. These compact representations can then be used at the server side for reconstruction or other tasks.

2.4 Generalizing to dataset shift

Wearable and sensor data is likely to undergo frequent dataset or covariate shifts (Quionero-Candela et al., 2009) - for example, when a sensor fails and is replaced by another one with slightly different sensitivity and zero error. Also, as environmental situations change, sensor readings may be affected. Inertial wearable sensors, for example, are sensitive to displacement effects when they are attached at slightly different positions every day (Baños et al., 2012). Thus, any method which aims to learn deep multimodal representations must be able to generalize in case the input domain shifts slightly or sensor readings fluctuate. In recent work, we show that *consistent* and *robust* representations help deep learning models to generalize better (I won the NeurIPS 2020 Competition on Predicting Generalization in Deep Learning, and will be presenting at NeurIPS this year). Building such consistent and robust representations is thus necessary, and methods can be explored to ensure that the generated representations are invariant to slight domain and environmental changes.

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